1. What is a class?
Class is a named group of properties and functions.
Classes can be used to create your own data type.

For example: If you want to create a data type which stores roll no., name and marks of students then you can create a
class which stores all these three instead of making three different arrays of these.

```
// Create a class
    class Student {
        int rollno;
        String name;
        float marks = 90;
    }
```

You can create its object like this:

```
Student[] abdul = new Student[5];
```

Now every object of Student class will contain rollno, name and marks.

2. Classes are not the real things they are just the templates or the logical instruct.
And if you want to create the real object you need to create instances of it.
Instaces are the things which occupies space in memory.
Like in above example "Student" is a data type or template which doesnot represent physical quantity whereas "abdul" is an instance of class "Student" and represents physical quantity.

3. Objects have 3 essential properties:
1. State of object:
The state or attributes are the built in characteristics or properties of an object.
For example, a T.V has the size, colour, model etc.

2. Identity of object:
It means whether one object is different from another.
Identity is identified using reference variable and if two reference variable are pointing to the same object in memory, they both are considered same.

3. Behaviour of the object:
The behavior or operations of an object are its predefined functions.

4. Instance Variable:
In class-based, object-oriented programming, an instance variable is a variable defined in a class (i.e. a member variable),
for which each instantiated(creation of object) object of the class has a separate copy, or instance.
An instance variable has similarities with a class variable, but is non-static.

5. "." opearator basically links reference variable to instance variable.

6. How to create objects of classes?

```
Student student1;  // Declaration
student1 = new Student(); // Creation
```

7. "new" keyword dynamically allocates memory and returns a reference to it.
Object will be created in heap memory and reference variable "student1" will be created in stack and "student1"
will be pointing to object in heap.

8. If the objects are not initialized then their default value is "null".

9. If we try to print instance variable of objects without initializing it then their default values or the values assigned
while declaration of object is printed.
For example: In the 2nd point if i try to print rollno, name and marks through instance "abdul" without initialising
it then the values printed will be (0, null, 90) where "0" and "null" are default values and "90" is the
value assigned while declaring instance variable.

10. A constructor is a special method that is automatically called when an object of a class is created.
To create a constructor, use the same name as the class, followed by parentheses ().

```
Student (int rollno, String name, int marks) {
this.rollno = rollno;
this.name = name;
this.marks = marks;
}
```

"this" will be replaces by reference name.
For example if you say:
Student abdul = new Student(1, "Abdul", 90);
then "this" will be replaced by "abdul".

11. It is a good convention to use "this".
If you say "rollno = rollno" instead of "this.rollno = rollno" then java will be unable to know which "rollno" is
representing to what and hence it will assign the roll no, the default value.

12. How to call one constructor from another?

```
Student () {
    // This is how you call one constructor from another constructor.
    // Internally it is something like this:
    // new Student(0, "Default person", 75.0f)
// Here, "this" is replaced with "Student".
    this (0, "Default person", 75.0f);
}
```

13. The this keyword refers to the current object in a method or constructor.

The most common use of the this keyword is to eliminate the confusion between class attributes and parameters with the same name.

Note:
We cannot access memory address in java and hence we cannot allocate memory manually, it happens dynamically.